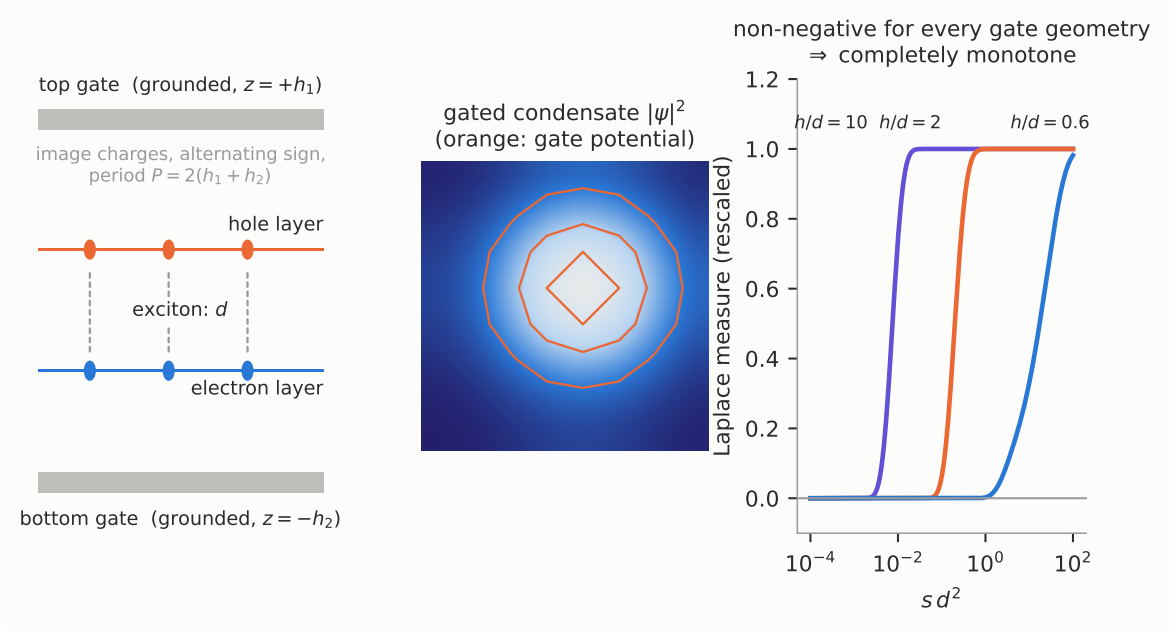


Gated exciton fluids:

a proof that image charges preserve complete monotonicity, a driven-dissipative solver test on rusty-SUNDIALS, and a laboratory protocol

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Abstract

A planar universal-optimality theorem for completely monotone interactions (OpenAI, September 2026; AI-generated, unrefereed) was used in a first note [1] as a dependency check on the exciton–exciton interaction of a 2026 bilayer experiment. That note left open whether the *dual-gated* device kernel, a sum of image charges of alternating sign, is completely monotone in r^2 (numerical evidence only), and whether gates could be engineered to break it. Here we **prove** (Theorem 1, paper and pencil, not in Lean) that for ideal grounded metal gates at arbitrary distances and arbitrary layer positions, the Laplace measure of the kernel in r^2 is a sum of manifestly non-negative terms, $S_m = 2\big(1 - \cos(b-a)\big)\big(1 + \cos(a+b)\big)$; the gates therefore cannot break complete monotonicity in this model. A by-product is the zero-wavevector coefficient $\tilde{U}(0) = 4\pi d\,(1 - d/D)$ for a gate separation D (a capacitor formula, not new, which fixes the experimental protocol). Second, we test the solver stack on the model the experiment's community would use for short-lived excitons or polaritons, a driven-dissipative Gross–Pitaevskii equation with an incoherent reservoir and a gate potential, in a pre-registered design with known answers: the steady states, the 768×768 linearisation (matched to 2.7×10^{-9}), the conservative limit and the IMEX threshold and order passed as registered; the stationarity and pump-budget gates of the gate sweep *failed* as registered at $t = 4000$ (residual up to 3.9×10^{-4}; the horizon was too short for the slow relaxation) and pass in an exploratory continuation to $t = 16000$, one seed per amplitude; the hypothesis that the emission shift lies between 0 and V_0 held at all three gate amplitudes. We report what failed or had to be changed (a registered cost was misjudged by two orders of magnitude and four amendments were written before the affected numbers existed). Third, we give a laboratory protocol with pass/fail criteria for the predictions that survive. We do not use Hugging Face materials data or neural-network surrogates, do not compare with data, and make no claim about exciton Bose–Einstein condensation. Written by an AI system; not refereed.

1 Why this note, and what it is not

The brief that prompted it asked for a driven-dissipative exciton fluid on rusty-SUNDIALS with materials data from Hugging Face. The first note [1] did something else (conservative gradient flows, a lattice theorem). We say so because the two are easily confused: that note did *not* contain a driven-dissipative model, an ARKode-type integrator, a gate, a defect, or any materials input. This one adds the first three in a minimal form (Section 3) and tests the ARKode-style module that exists in the library; it does not add the last (Section 5).

What is established here. (i) Theorem 1 below, a short exact argument (checked numerically on 300 random geometries). (ii) Registered numerical gates on a standard open-dissipative model: which passed, which failed, in Section 4. **What is not.** No quantum phase transition is located; no experiment is reproduced; the model parameters are dimensionless and not those of any device; the protocol of Section 6 is a proposal that nobody has run.

2 Gated kernels are completely monotone

Setting. Grounded perfect-conductor planes at $z = +h_1$ and $z = -h_2$; a homogeneous dielectric between them (a uniform ε is a prefactor, an anisotropic one rescales z by a constant); an electron layer at z_e and a hole layer at $z_h = z_e + d$. The exciton–exciton kernel is $U = V_{ee} + V_{hh} - 2V_{eh}$, V the Green function of the box. With $P = 2(h_1 + h_2)$ a unit charge at z_0 produces at (z, r)

$$V(z, z_0; r) = \sum_{n\in\mathbb{Z}} \Big[\big(r^2 + (z - z_0 - Pn)^2\big)^{-1/2} - \big(r^2 + (z - 2h_1 + z_0 - Pn)^2\big)^{-1/2} \Big].$$

Theorem 1. *U is completely monotone as a function of $t = r^2$ on $(0, \infty)$: $U = \int_0^\infty e^{-st}\mu(s)\,ds$ with $\mu \geq 0$.*

Proof. Write each term as $(t + a^2)^{-1/2} = \pi^{-1/2}\int_0^\infty e^{-s(t+a^2)}s^{-1/2}ds$. Then $\mu(s) = (\pi s)^{-1/2}S(s)$ where $S(s)$ is the sum over images of e^{-sa^2} with the signs of $V_{ee} + V_{hh} - 2V_{eh}$. Poisson summation, $\sum_n e^{-s(c+Pn)^2} = (\pi/s)^{1/2}P^{-1}\sum_{m\in\mathbb{Z}} e^{-\pi^2m^2/(P^2s)}\cos(2\pi mc/P)$, gives $S(s) = (\pi/s)^{1/2}P^{-1}\sum_m e^{-\pi^2m^2/(P^2s)}S_m$ with, for $\alpha = 2\pi m/P$, $a = \alpha(z_e - h_1)$, $b = \alpha(z_h - h_1)$ (so $\alpha d = b - a$),

$$S_m = 2 - \cos 2a - \cos 2b - 2\cos(b - a) + 2\cos(a + b) = 2\big(1 - \cos(b - a)\big)\big(1 + \cos(a + b)\big) \geq 0,$$

using $\cos 2a + \cos 2b = 2\cos(a + b)\cos(a - b)$; and $S_0 = 0$ by neutrality. Hence $S(s) \geq 0$ termwise. □

Consequences, and limits of the claim. The single-gate and gate-free limits are pointwise limits of completely monotone functions and so are completely monotone. Within *this* model the gates cannot be engineered to violate the hypothesis of the lattice theorem: Kagome or stripe ground states, if observed in a gated device, must come from something the model leaves out, which includes dielectric image charges of weight κ^n with $|\kappa| < 1$ (the Poisson form is then not a pure cosine series; *not proved, not tested*), the finite screening of real gates, disorder, and quantum fluctuations. The lattice theorem itself is imported and was not refereed. A Lean formalisation needs the Poisson summation of Mathlib's Jacobi theta; we did not do it. The zero-wavevector coefficient follows from the same series (and from Gauss's law for three capacitors in series):

$$\tilde{U}(0) = \int U\, d^2r = 4\pi d\left(1 - \frac{d}{D}\right), \qquad D = h_1 + h_2. \tag{1}$$

We checked it against the numerical integral of the image series (agreement to 3×10^{-3} relative at $D = 3d$ and $D = 20d$, limited by the integration cutoff). It is a capacitor formula, textbook; we use it because it is a quantitative, device-geometry-dependent prediction (Section 6).

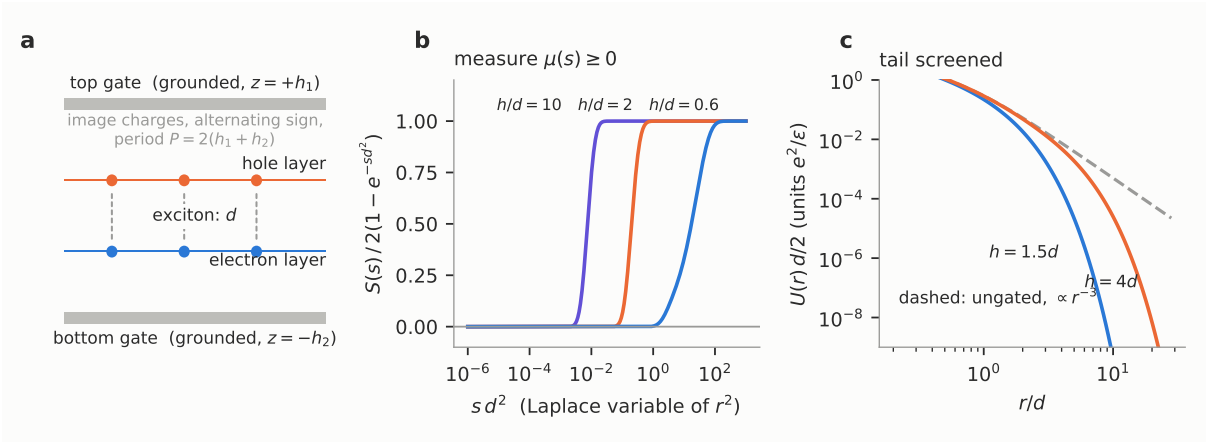


Figure 1: (a) The device and its image series. (b) $S(s)/2(1 - e^{-sd^2})$ from the exact Poisson form for three symmetric gate distances: non-negative everywhere, as Theorem 1 states. (c) The kernel with and without gates: the gates remove the dipolar r^{-3} tail (exponential screening at $r \gg h$) without leaving the completely monotone class. All three panels are generated from the formulas of the text by `phase3_figures.py`.

3 A driven-dissipative model with a gate, tested on rusty-SUNDIALS

Model. Dimensionless units $\hbar = m = 1$,

$$i\,\partial_t \psi = \Big[-\tfrac{1}{2}\nabla^2 + V(\mathbf{x}) + g|\psi|^2 + g_R n_R + \tfrac{i}{2}\big(R n_R - \gamma\big)\Big]\psi, \tag{2}$$

$$\partial_t n_R = P(\mathbf{x}) - (\gamma_R + R|\psi|^2) n_R, \tag{3}$$

the standard open-dissipative (Wouters–Carusotto) form with $g = 0.04$, $g_R = 1$, $\gamma = 0.1$, $\gamma_R = 200$, $R = 1$ (stiff ratio $\gamma_R/\gamma = 2000$), uniform pump P and gate $V = V_0 e^{-d^2/2\sigma^2}$. Above threshold $P_{th} = \gamma\gamma_R/R = 20$ the uniform state is $n = P/\gamma - \gamma_R/R$, $n_R = \gamma/R$, $\omega = gn + g_R n_R + V$ (we use $P = 21$: $n = 10$, $n_R = 0.1$, $\omega = 0.5$). The model is generic; it is the model one would write for short-lived excitons or polaritons. It is *not* the model of the equilibrium experiment of Qi *et al.*, whose dipolar excitons live microseconds.

Design. Pre-registered in `docs/designs/EXCITON_FLUID_PHASE3_PREREG.md` (commit `335caa5`, before any run of the Rust program), independent Python reference first (`phase3_reference.py`). Gates: D-a steady states above and below threshold; D-b the 768×768 linearisation of the solver against the reference 3×3 per-k spectrum; D-c conservative limit (norm and energy); D-d gate sweep ($V_0 \in \{0, 0.25, 0.5, 1\}$): pump budget $\sum P = \sum(\gamma_R n_R + \gamma|\psi|^2)$, single locked frequency, and the hypothesis $0 < \omega(V_0) - \omega(0) < V_0$; D-e the ARKode-style IMEX module, whose implicit stage is a fixed-point iteration and so must fail exactly when $h\gamma_R a_{ii} \geq 1$; D-f its temporal order; D-g a planted bug. A deliberately *wrong* prediction is excluded by construction: the intuitive local-density guess that the gate leaves the density pinned at $P/\gamma - \gamma_R/R$ is incompatible with a single locked frequency (the gate shifts the local frequency, so currents must flow), and was not registered.

Amendments, all before the numbers they affect. (A4, on the number of seeds of D-g, is in the pre-registration.) A1: the IMEX module's fixed-point test squares the tolerance (effective absolute tolerance rtol (atol + rtol$|z|$)), so 10^{-12} cannot converge in floating point; set to 10^{-6}.

A2: integrating D-a/D-d/D-g in the laboratory frame needed more than 23 minutes for one run (the phase rotates at $\omega = 0.5$ and BDF must resolve it for $t = 4000$); the flow is integrated in the frame rotating at the registered value of ω with all gate quantities evaluated gauge-covariantly. A3: even then the pinned CVODE refactors a dense matrix every few steps (43 ms per step at 768 unknowns, hours per run, about 70 runs), so **D-a, D-d, D-g were run on an 8×8 grid with** $\sigma = 1.5$ (same σ/L), not 16×16; D-b, D-c, D-e, D-f ran on 16×16 as registered. The consequence is that the gate-sweep results describe a very small box.

4 Results of the registered gates

gate	grid	outcome (numbers from `summary.json`)	registered result
D-a	8×8	above threshold, 10 seeds: n, n_R, ω within 1.2×10^{-12}, 5.9×10^{-14}, 4.8×10^{-13} of the closed form; below: $\max \lvert\psi\rvert \leq 1.6 \times 10^{-11}$	pass
D-b	16×16	eigenvalue multiset of the solver's 768×768 Jacobian equals the reference to 2.7×10^{-9}; largest real part 1.4×10^{-10} (the Goldstone mode)	pass
D-c	16×16	norm drift 1.6×10^{-11}, energy drift 4.1×10^{-10} over $t = 20$ (6464 steps)	pass
D-d (i),(ii)	8×8	pump budget up to 5.0×10^{-7} (gate 10^{-8}), stationarity residual up to 3.9×10^{-4} (gate 10^{-7}); even $V_0 = 0$ gives 1.6×10^{-7}	**fail**
D-d (iii)	8×8	$\omega(V_0) - \omega(0) = 0.0488, 0.0876, 0.1422$ for $V_0 = 0.25, 0.5, 1$: between 0 and V_0, increasing	pass (hypothesis)
D-e	16×16	one step succeeds at $h = 0.005, 0.01, 0.0125$ and returns an error at $0.02, 0.05$ (threshold 0.01707); no silent non-finite result; trajectory to $t = 20$ within 7.3×10^{-6} of CVODE	pass
D-f	16×16	error ratios 4.20 and 4.07 on halving h (second order)	pass
D-g	8×8	planted bug: $n = 220.0$ instead of 10; budget violated by ≥ 0.52 (3 seeds, amendment A4)	pass

Table 1: Gates as registered. Every number is generated from the run records by `phase3_paper_numbers.py`.

The registered gate-sweep gates failed. At $t = 4000$ the stationarity residual of the gated states is up to 3.9×10^{-4} and the pump budget is violated by up to 5.0×10^{-7}, against thresholds 10^{-7} and 10^{-8}; the ungated state also exceeds its threshold (1.6×10^{-7}). The horizon was set from the decay rate of the ungated Bogoliubov modes (2.4×10^{-3}) and was too short: the gated state relaxes more slowly. All ten seeds end on the same state to all printed digits, and the residual does not depend on the integrator tolerance (rtol $= 10^{-8}$ to 10^{-12} give 7.6×10^{-5}), so it is not integration error. *Exploratory* continuations (not gates, one seed each) to $t = 16000$ give residual 1.0×10^{-8} and budget violation 1.1×10^{-13} for $V_0 = 0.5$, which would pass; the other amplitudes give $V_0 = 0$: residual 3.3×10^{-10}, budget 0.0×10^{0}; $V_0 = 0.25$: residual 6.0×10^{-8}, budget 2.4×10^{-14}; $V_0 = 1$: residual 8.7×10^{-9}, budget 6.9×10^{-13} (all within the registered thresholds).

What the gate sweep shows, and does not show. In the small box the gate depletes the condensate at its centre (from 10 to 7.07, 4.58, 1.49 for $V_0 = 0.25, 0.5, 1$, while the corner rises to 11.08, 11.92, 13.06) and the single emission frequency moves by 0.0488, 0.0876, 0.1422, i.e. by 0.20, 0.18 and 0.14 of V_0, decreasing with V_0 as the condensate leaves the gate region. The

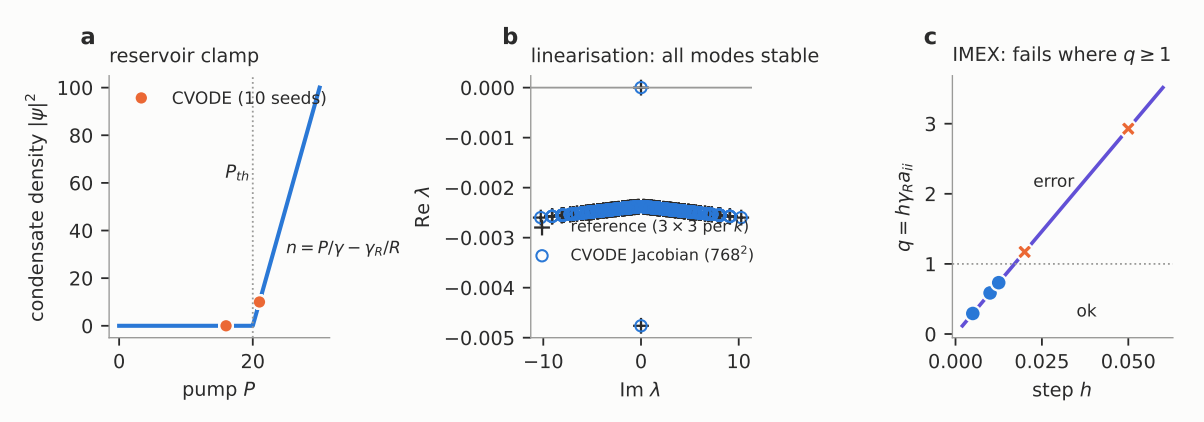


Figure 2: Solver tests. (a) Steady-state density against pump: closed form (line) and ten CVODE runs per pump (dots; they coincide to 1.2×10^{-12}). (b) The eigenvalues of the solver's finite-difference Jacobian (circles) lie on the reference spectrum (crosses); the isolated points are the $k = 0$ phase (Goldstone) and amplitude modes; the fast reservoir mode near $-\gamma_R$ is outside the plotted range. (c) The IMEX module's fixed-point stage: contraction factor $q = h\gamma_R a_{ii}$; it returns an error exactly where $q > 1$.

linearisation about the numerical states has no unstable mode within the (relaxation-limited) accuracy 9.2×10^{-6}. These are statements about an 8×8 periodic box with a gate width comparable to the healing length; no finite-size study was made, and the dependence on the gate width and on the pump was not scanned.

The ARKode-style module. It does what its design says: a fixed-point implicit stage that converges iff $h\gamma_R a_{ii} < 1$, so for the stiff reservoir of this model ($\gamma_R = 200$) the step is limited to 0.017 while CVODE's BDF takes the full integration; with h below the threshold it is second order and agrees with CVODE to 7.3×10^{-6}. It is not an adaptive stiff integrator and the library's default `ImexSolver` should not be used for stiff problems without the replacement of the implicit stage.

Cost. The pinned CVODE factorises a dense matrix every few steps (43 ms per step at 768 unknowns, 1.4 s per $t = 20$ at 192 unknowns); this, not the physics, set the grid of D-d. A Krylov or banded implicit solver is the obvious requirement for a 64×64 grid and was not available in the pinned tree.

5 What was not done

Materials data and surrogates (Hugging Face, GNNs). Not used. The gate is a Gaussian of chosen amplitude, the parameters are dimensionless. Predicting exciton binding energies from a graph neural network trained on bulk or monolayer data for a twisted bilayer with defects would require a validity check that has not been made (the training distributions of MACE/ALIGNN-type models are not heterostructures with interlayer excitons); the output of such a model would be an input parameter with an uncontrolled error, not a result.

The Wigner transition. The classical statement of the lattice theorem is a zero-temperature, zero-fluctuation statement; where the quantum fluid freezes needs quantum Monte Carlo (the literature on two-dimensional dipolar bosons has such calculations; we did not read them in this

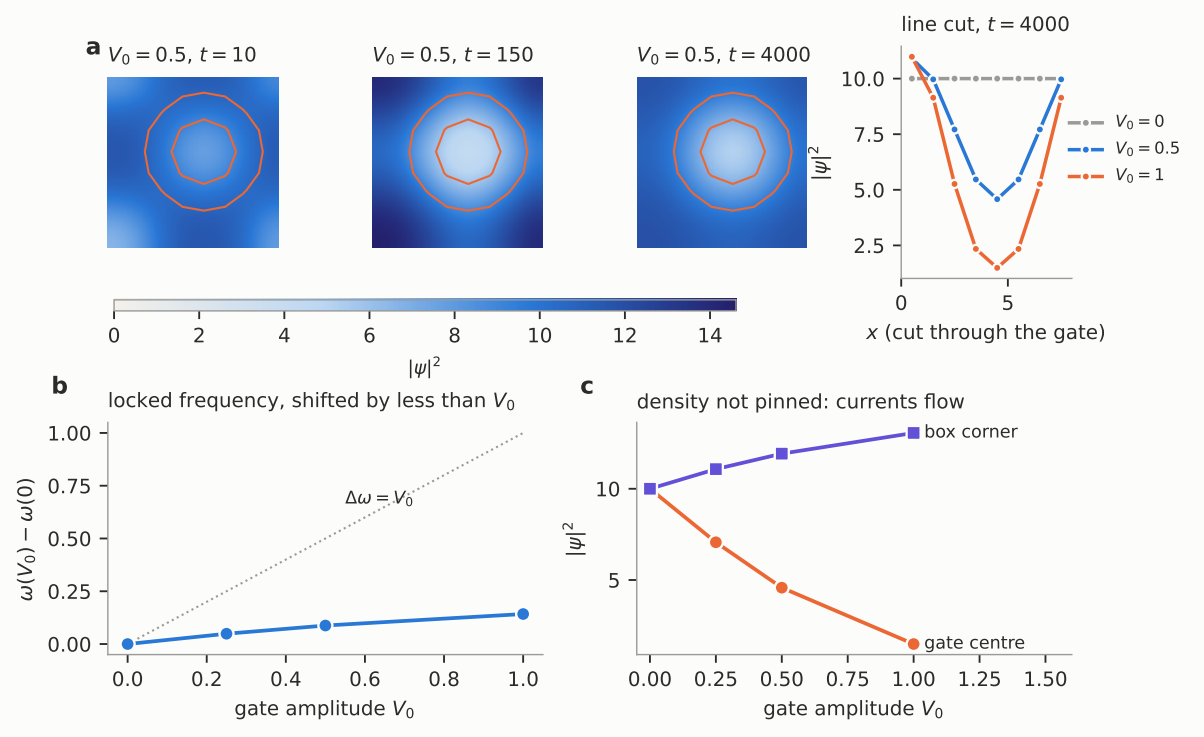


Figure 3: A gated driven-dissipative condensate on the 8×8 grid ($\sigma = 1.5$), uniform pump $P = 21$. (a) Density from the uniform start with noise at three times (orange: contours of the gate; fields are bicubic interpolations of the 8×8 values) and a line cut through the gate at $t = 4000$ for three amplitudes. (b) The locked emission frequency shifts with the gate amplitude by less than V_0. (c) Density at the gate centre falls and at the box corner rises: the density is *not* pinned at its ungated value, because the single locked frequency requires currents (largest current density 7.6×10^{-3}–1.3×10^{-2} for $V_0 = 0.25$–0.5).

session and quote no number). What can be computed with the present tools is a harmonic-phonon (Lindemann) estimate for the gated kernel from the Hessian code of the first note: a crude order-of-magnitude, to be labelled as such and compared with the QMC literature before use. Not done.

Frustration and edges. The first note found frustration energies of 7.9×10^{-7}–8.6×10^{-3} above the lattice energy on incommensurate tori. Open boundaries, domain walls and edge currents are not in that note and not here. A design (disks of N particles, boundary shapes, count of 5–7 disclinations against the registered lattice energy) is a straightforward extension of the first note's N-particle flows; it is the cheapest next numerical study and has not been started.

6 Laboratory protocol

The aim is to turn the statements that survived into measurements that can fail. Each item gives the quantity, the prediction, the pass/fail rule and the confounders. None has been performed; the model parameters of Section 3 must be *measured* on the device (items L1–L3) before the later items are meaningful.

Platform and regime. A dual-gated transition-metal-dichalcogenide bilayer or heterobilayer with hBN spacers, graphite (or metal) top and bottom gates, device series with gate separations

D spanning at least $5d$ to $50d$ (d the electron–hole layer separation), one-gate and two-gate versions. L4 and L5 need a regime in which the dipolar excitons are cold and degenerate or crystalline (T below the sample's melting or degeneracy temperature, to be established by L2); L6 needs a driven regime with excitons or polaritons of lifetime short compared with the observation time (a microcavity, or intralayer excitons under continuous pumping). The two regimes are different samples or different conditions; the model of Section 3 does not apply to long-lived interlayer excitons.

Common controls. Gates grounded and floating; a device without the second gate; a detuned pump; temperature series; at least three devices per geometry; blind analysis (fits fixed before the data of the discriminating series are read); raw data and analysis code deposited with the preregistration of the series. Density is calibrated independently of the quantity tested (capacitance or absorption), not from the photoluminescence blueshift itself.

	Measurement	**Prediction (from this note)**	**Pass / fail, confounders**
L1	Interaction strength: PL peak shift versus calibrated exciton density n, device series in D, low T, below degeneracy.	$dE/dn = \tilde{U}(0) = (4\pi e^2 d/\varepsilon)\,(1 - d/D)$ (Eq. 1): increases with D and saturates at the ungated value. Dielectric mismatch and finite gate screening are outside the model, so the *trend* is the test.	Fit slope vs D per device; fail if the slope is independent of D within 10% over $D/d \in [3, 20]$, or has the opposite trend. Confounders: exchange and correlation corrections (a D-independent offset), density calibration, heating.
L2	Phase diagram: structure factor $S(\mathbf{k})$ (momentum-resolved emission or interferometry) versus n and T at fixed D.	No claim of a number. The classical lattice has a hexagonal Bragg ring at $k_0 = 4\pi/(\sqrt{3}a)$ with $n = 2/(\sqrt{3}a^2)$, a its constant.	Establishes where L3 applies (fluid, solid). A ring of the wrong k_0 for the measured n is a calibration failure, not a result.
L3	Lattice symmetry versus gate geometry in the solid regime: classify peaks of $S(\mathbf{k})$ (six-fold or other) over the series in D, one and two gates.	Within the ideal-metal model, Theorem 1 and the lattice theorem (conditional) give a triangular lattice at every D in the classical limit.	A non-hexagonal pattern at some D is *not* a refutation of the theorem (quantum fluctuations, disorder, finite size and non-ideal gates are outside it); it is evidence for one of those, to be separated by the temperature and disorder series. A hexagonal pattern at all D is the null result.
L4	Gate-voltage tuning of the local emission energy: nanofabricated finger gate, DC bias V_g, pump fixed above condensation threshold, spatially resolved spectroscopy.	Locked single frequency; shift $\Delta\omega$ increasing with the gate amplitude but smaller than the local energy shift (Section 4; qualitative, small-box numerics).	Fail if the line splits into two frequencies at the gate centre and outside at all V_g (no locking) or if $\Delta\omega$ exceeds the independently measured local Stark shift. Confounders: Stark shift calibration, heating at the gate.

	Measurement	Prediction (from this note)	Pass / fail, confounders
L5	Threshold and clamp: condensate-peak intensity and reservoir (high-energy tail) intensity versus pump P.	Condensate density linear in P above P_{th} with slope $1/\gamma$; reservoir density clamped at γ/R (Section 3).	Fail if reservoir emission keeps growing linearly above P_{th} within error. Confounders: detection saturation, multimode condensation, thermal tail.
L6	Edge/frustration (exploratory): boundary-shaped trap arrays with N excitons.	No numerical prediction yet (Section 5).	Do not run before the numerical design exists.

Decision rule. L1 and L5 are the first experiments: they need no new fabrication beyond a device series and they can fail cleanly. A failure of L1's trend invalidates the use of ideal-metal image kernels for that platform and with it the applicability of Theorem 1 there. L2 must come before L3 and L4; without L2 a lattice claim is unfalsifiable.

7 Limitations

- Theorem 1 is for ideal grounded metals and a homogeneous dielectric; it is not in Lean; the lattice theorem it feeds is unrefereed and conditional in the first note's Lean files.
- The numerical model is generic and dimensionless; one parameter set; D-a, D-d, D-g on an 8×8 grid; no finite-size or resolution study; the solver (dense CVODE, ≤ 768 unknowns) is not claimed to be the best of its kind.
- The ARKode-style module tested is a fixed-step ARS222 with a fixed-point implicit stage; it is not SUNDIALS' ARKode. Its failure at $h\gamma_R a_{ii} \geq 1$ is its design, not a defect of the library; a Newton–Krylov implicit stage would be the replacement.
- No experimental data; no Hugging Face data; no contact with any experimental group; written by an AI system and not reviewed by a human.

Reproducibility

Pre-registration, amendments, run records and summary: `docs/designs/EXCITON_FLUID_PHASE3_PREREG.md`, `exploration/exciton/results/phase3/`; instrument `exploration/exciton/rust/qf-exciton-p1/src/bin/dd.rs` (rusty-SUNDIALS tag v11.6.0; `-features m8` for the 8×8 runs); reference `phase3_reference.py`; gate evaluator `phase3_summarise.py`; the numbers of this note `phase3_paper_numbers.py`; figures `phase3_figures.py`; proof check `gate_image_cm_*.py`. First note: concept DOI 10.5281/zenodo.23289026.

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